CLINICAL · NOSE & SINUSES

Deviated septum, septal haematoma and abscess

The septum is the midline wall that holds up the lower nose and divides the airway. A deviated septum only matters when it blocks the patient. A swollen septum after a blow is the emergency: a haematoma drained the same day is a minor procedure, while one left alone becomes an abscess, a dead cartilage and a saddle nose.

AT A GLANCE

- Treat deviation only if it causes symptoms.
- Always examine the septum after trauma.
- Drain a haematoma the same day.
- Keep an L-strut in septoplasty.
- Large perforations can now be closed with an anterior ethmoidal artery flap; buttons are no longer a routine first step.

1 CHAPTER ONE Overview

The partition of the nose.

1.1 Applied anatomy

The septum has three parts: the *columellar septum* (the columella, with the medial crura of the alar cartilages), the *membranous septum* (skin between the columella and the septal cartilage), and the *septum proper*, made of cartilage and bone covered by mucoperichondrium and mucoperiosteum.

1.1.1 The framework

- **Quadrilateral (septal) cartilage:** the front of the septum, supporting the nasal tip and dorsum.
- **Perpendicular plate of the ethmoid:** above and behind.
- **Vomer:** below and behind.
- **Small contributions:** the crest of the maxilla and palatine bones, the nasal spine of the frontal bone, and the rostrum of the sphenoid.

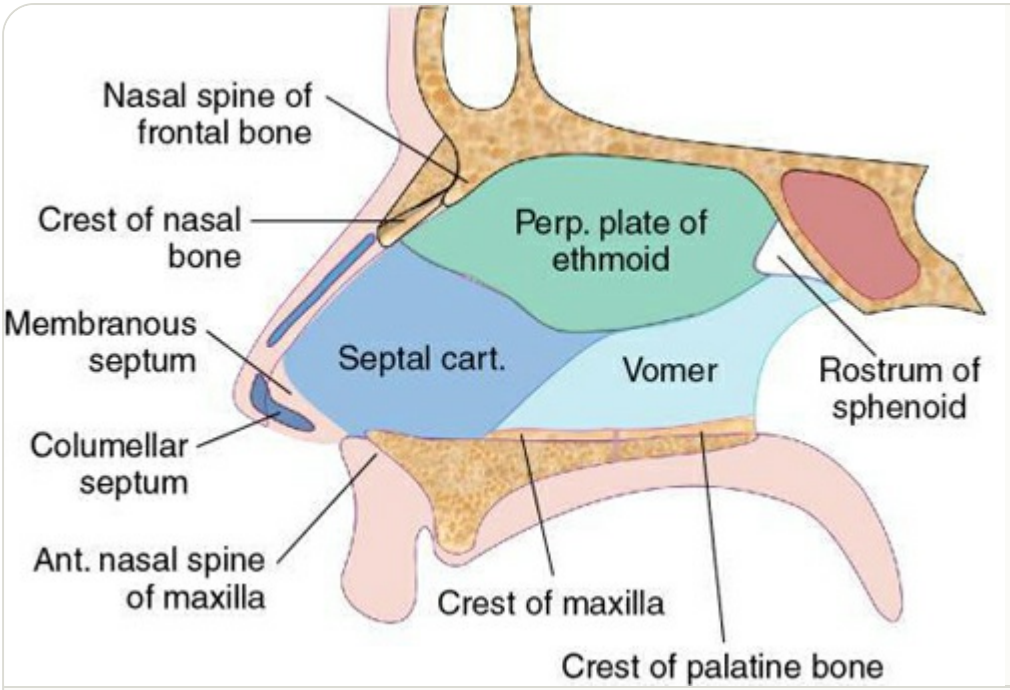


FIGURE 1 Anatomy of the nasal septum: septal cartilage, perpendicular plate of the ethmoid and vomer.

Dhingra, fig. 26.1

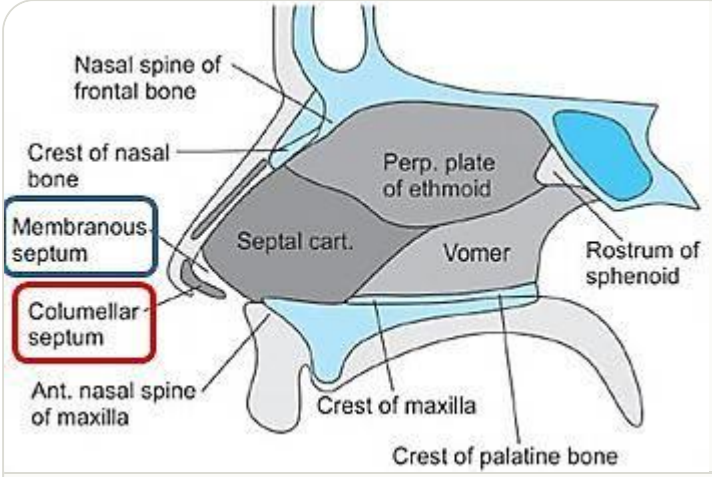


FIGURE 2 The septal cartilage and bones of the septum.

Anatomy atlas · ENT Steps p. 104

1.1.2 Blood supply

Little's area on the anteroinferior septum, where four arteries meet, is the commonest site of epistaxis (see the epistaxis topic).

1.2 Why it matters

Septal problems cause nasal obstruction, sinusitis, epistaxis and external

deformity, and septal surgery is one of the commonest operations in ENT.

2 CHAPTER TWO Septal trauma

Fractures of the septum.

2.1 How it happens

Blows from the front, side or below buckle, fracture, dislocate or crush the septum. The septal cartilage can dislocate from the vomerine groove without any fracture of the nasal bones. Mucosal tears cause profuse epistaxis; an intact mucosa traps blood as a haematoma.

2.2 Named fractures

- **Jarjaway fracture:** from a frontal blow, running horizontally backwards just above the junction of the septal cartilage with the vomer.
- **Chevallet fracture:** from a blow from below, running vertically from the anterior nasal spine up to the junction of the bony and cartilaginous dorsum.

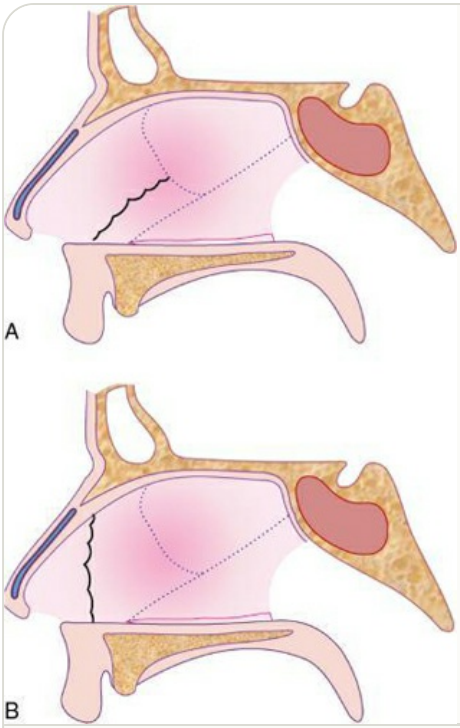


FIGURE 3 Septal fractures: (A) Jarjaway type, (B) Chevallet type.
Dhingra, fig. 26.2

2.3 Treatment

Drain any haematoma; reposition dislocated or fractured fragments, support them between the flaps with mattress sutures and packing, and treat any nasal bone fracture at the same time. Ignored septal injuries leave a deviated cartilaginous nose and an asymmetric tip, columella or nostril.

CHAPTER THREE

3 Deviated nasal septum

When the partition is crooked.

3.1 Causes

3.1.1 Trauma

A lateral blow dislocates the cartilage from the vomerine groove and maxillary crest; a frontal blow buckles and telescopes it. Childhood injuries, and birth trauma from a difficult delivery, are often forgotten.

3.1.2 Developmental

Unequal growth of the palate and the skull base buckles the septum. Mouth breathers with adenoid hypertrophy often have a high-arched palate and a deviated septum; it is also seen with cleft lip and palate and dental anomalies.

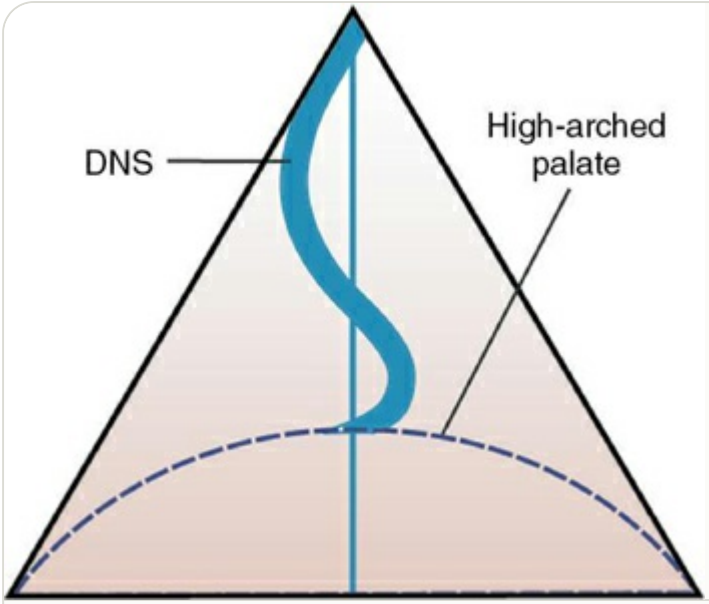


FIGURE 4 Deviated septum with a high-arched palate.
Dhingra, fig. 26.3

3.1.3 Racial and hereditary

Commoner in white populations, and often in several members of a family.

3.2 Types

- **Anterior (caudal) dislocation:** the lower border of the cartilage projects into one nostril, best seen with the head tilted back.
- **C-shaped deviation:** a single curve; the wider side often shows compensatory turbinate hypertrophy.
- **S-shaped deviation:** two curves, which can block both sides.
- **Spur:** a shelf-like projection, usually at the junction of bone and cartilage, which may press on the lateral wall.
- **Thickening:** from an organised haematoma or overlapping fragments.

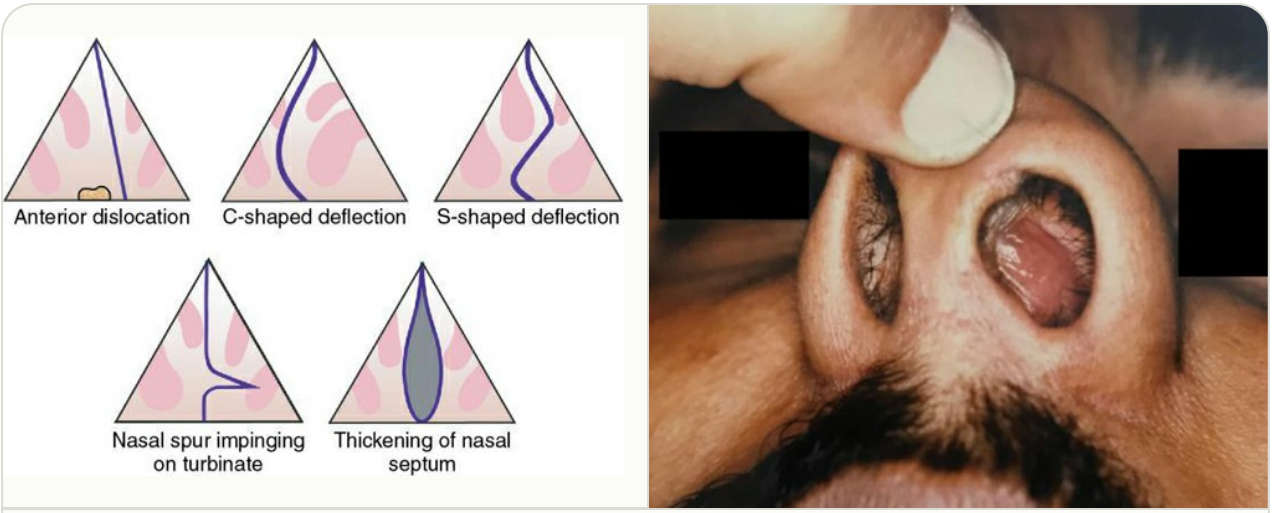


FIGURE 5 First, the types of septal deviation. Second, a C-shaped deviation nearly blocking the left side.

Dhingra, figs 26.4 and 26.6

3.3 Clinical features

3.3.1 Nasal obstruction

One-sided or both; high deviations block more than low ones, because the main airflow passes through the upper nose. Paradoxical obstruction on the wider side comes from compensatory turbinate hypertrophy.

3.3.2 Other symptoms

- **Headache,** from a spur pressing on the lateral wall.
- **Sinusitis,** from obstructed ostia and poor ventilation.
- **Epistaxis,** from crusting over the convexity or vessels stretched over a spur.
- **Reduced smell,** when air cannot reach the olfactory cleft.
- **External deformity** of the dorsum, tip or columella.
- **Middle ear infections.**

3.4 Where is the obstruction?

LEVELS OF NASAL OBSTRUCTION

LEVEL	CAUSES
Vestibule	Caudal septal dislocation, stenosis, synechiae

LEVEL	CAUSES
Nasal valve	Narrow valve angle, collapse after rhinoplasty, synechiae
Attic (upper septum)	High septal deviation
Turbinates	Hypertrophy, concha bullosa
Choana	Choanal atresia, choanal polyp, adenoids

3.5 The nasal valve

The narrowest part of the airway, bounded by the septum, the lower edge of the upper lateral cartilage (normally at an angle of about 10 to 15 degrees) and the head of the inferior turbinate. *Cottle's test*: pulling the cheek laterally opens the valve; if breathing improves, the valve is part of the problem.

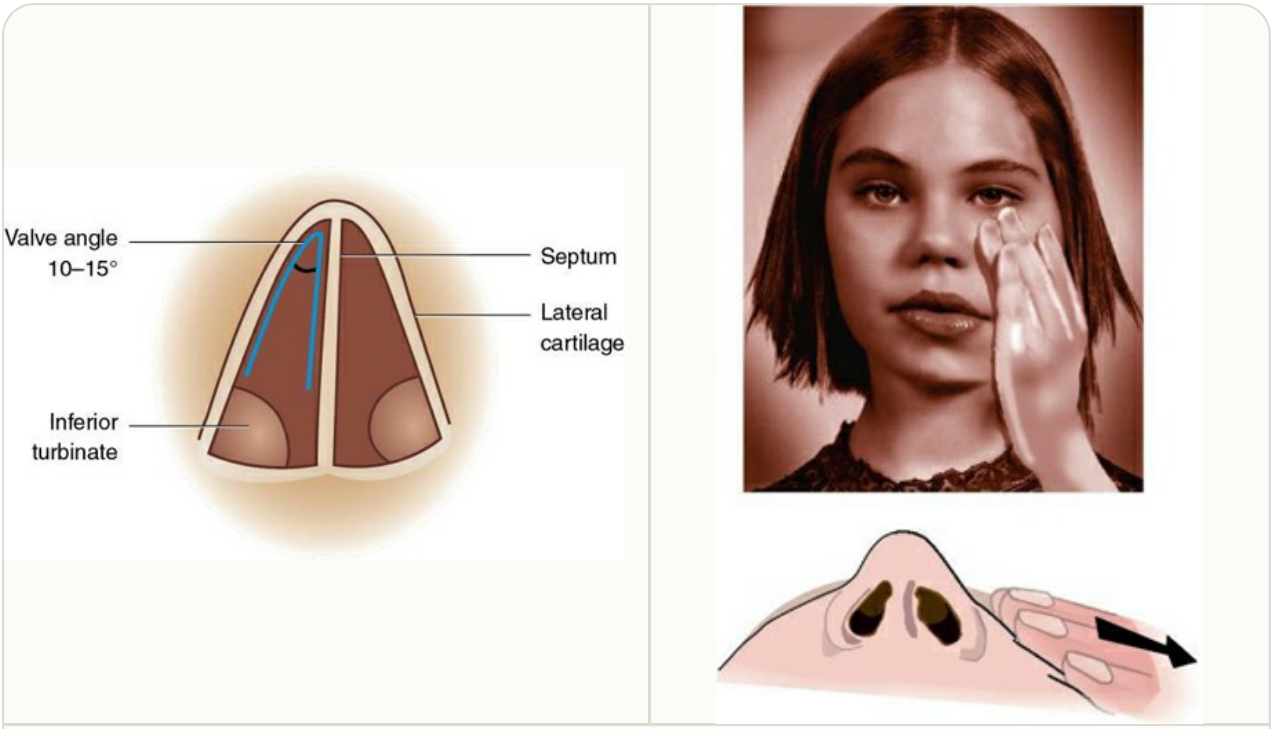


FIGURE 6 First, the nasal valve area. Second, Cottle's test: pulling the cheek laterally opens the valve.

Dhingra, figs 23.3 and 26.7

3.6 Management

Minor, symptomless deviations need nothing. Treat coexisting rhinitis first. Operate when the deviation causes obstruction or the problems listed above.

3.6.1 Septoplasty

The conservative operation: only the most deviated parts are removed; the rest is straightened (by scoring, morselising or repositioning) and put back. A flap is usually raised on one side only. It has largely replaced submucous resection, and can be done in children with severe obstruction because it preserves growth centres.

3.6.2 Submucous resection

Flaps are raised on both sides and the deviated cartilage and bone are removed, leaving dorsal and caudal struts; now mostly historical (see the nose and sinus operations topic).

3.6.3 Associated procedures

Turbinate reduction for hypertrophy, and septorhinoplasty when the external nose is deviated.

OPERATIVE GUIDES IN THEATRE 4 operations

Septoplasty—Classic and Endoscopic
Septoplasty
Nasal Septal Perforation
Perforation
Surgery of the Turbinate
Turbinate surgery
Rhinoplasty and Surgery of the Nasal Valve
Nasal valve work

4 CHAPTER FOUR Septal haematoma and abscess

The emergency.

4.1 Septal haematoma

4.1.1 Cause

Blood under the perichondrium or periosteum, after nasal trauma or septal surgery, or spontaneously in bleeding disorders.

4.1.2 Clinical picture

Bilateral nasal obstruction, frontal headache and pressure over the bridge, with a smooth, rounded, soft swelling of the septum in both nostrils.

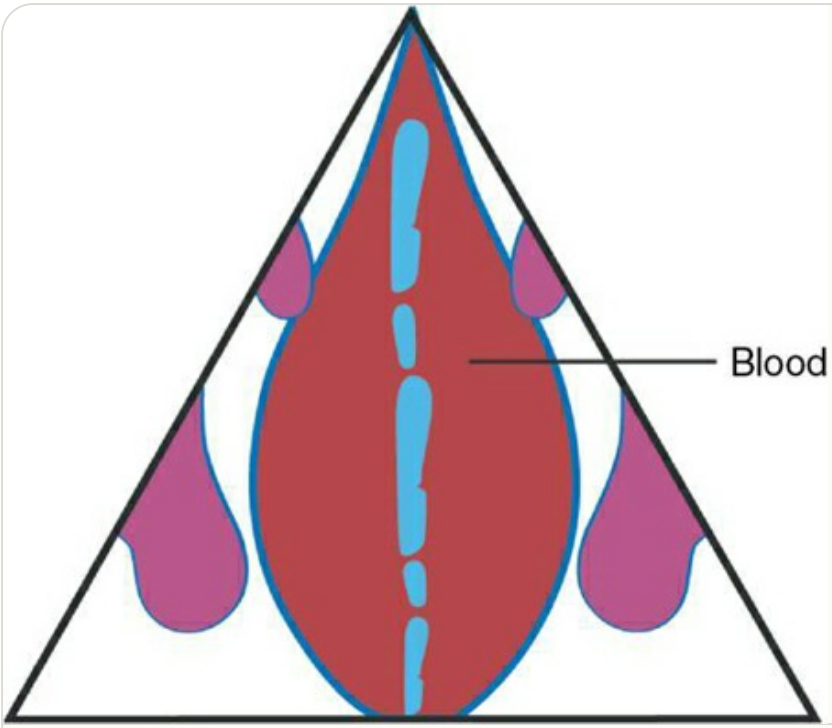


FIGURE 7 Septal haematoma: smooth swellings of the septum in both nostrils.
Dhingra, fig. 26.8

4.1.3 Treatment

Aspirate small collections with a wide-bore needle; incise larger ones parallel to the floor, excise a small strip of mucosa to keep it draining, pack both sides to prevent reaccumulation, and give antibiotics to prevent abscess.

4.2 Septal abscess

4.2.1 Cause

Usually an infected haematoma; occasionally after a furuncle of the nose or lip, or systemic infections such as typhoid or measles.

4.2.2 Clinical picture

Severe obstruction, pain and tenderness over the bridge, fever and chills, a red, swollen nose, fluctuant septal swelling and tender submandibular nodes.



FIGURE 8 Septal abscess.
Dhingra, fig. 26.9

4.2.3 Treatment

Drain urgently at the most dependent point, excise a piece of mucosa, remove pus and dead cartilage, reopen daily for two or three days if needed, and give antibiotics guided by culture for at least ten days.

4.3 Complications

- Cartilage necrosis and *saddle nose* (supratip depression), corrected by augmentation rhinoplasty after two or three months.
- Septal perforation.
- Meningitis and cavernous sinus thrombosis, rare but serious.

NEVER SEND HOME A SWOLLEN SEPTUM

A child or adult with nasal obstruction after a blow needs the septum examined. A missed haematoma in a child may only show itself years later as a saddle nose.

5 CHAPTER FIVE Septal perforation

A hole in the partition.

5.1 Causes

CAUSES OF SEPTAL PERFORATION

- TRAUMA **Septal surgery** repeated cautery, nose picking, piercing
- INFECTION AND INFLAMMATION **Septal abscess** myiasis, rhinolith or foreign body, tuberculosis, lupus, leprosy, syphilis
- DESTRUCTIVE DISEASE **Granulomatosis with polyangiitis** NK/T-cell lymphoma, carcinoma
- DRUGS AND CHEMICALS **Cocaine** long-term steroid sprays, chromium, arsenic, soda ash
- UNKNOWN **Idiopathic**

Lupus, tuberculosis and leprosy perforate the cartilaginous septum; syphilis perforates the bony septum.

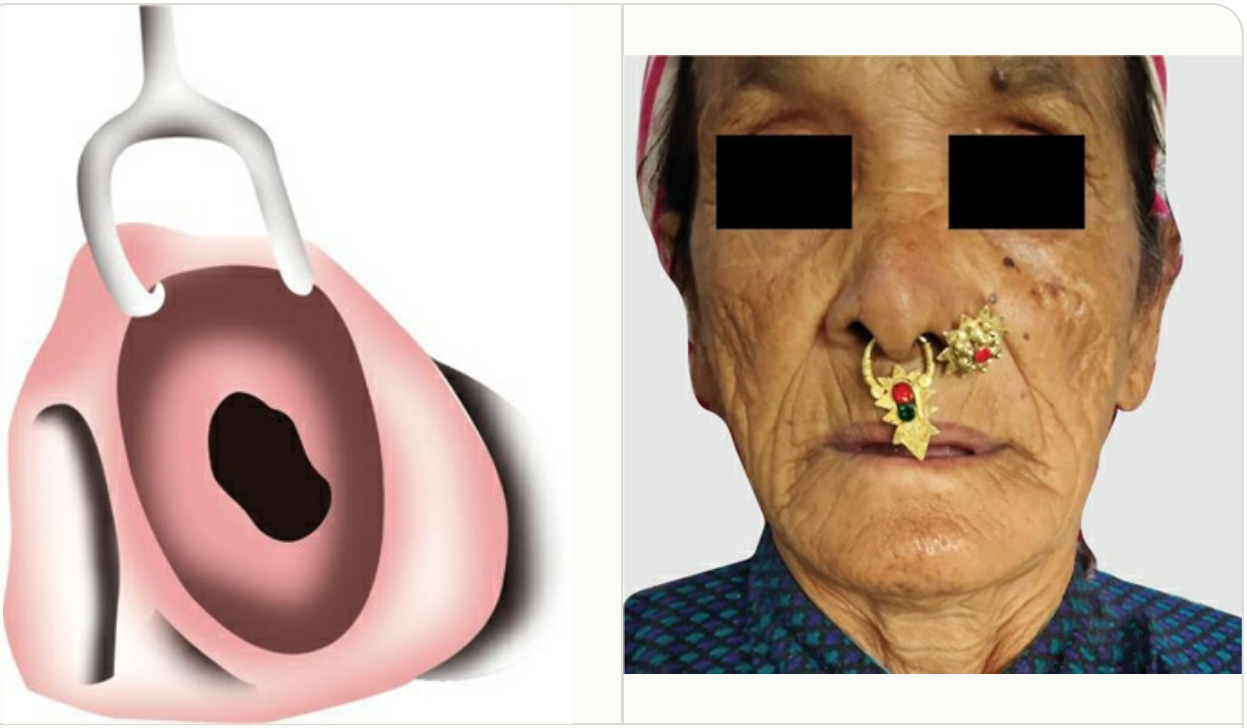


FIGURE 9 First, a septal perforation. Second, a septum deliberately perforated for an ornament.
Dhingra, figs 26.10 and 26.11

5.2 Clinical picture

Small anterior perforations whistle with breathing. Larger ones crust, obstruct and bleed when crusts are removed; very large ones can cause a saddle nose.

5.3 Investigations

Look for the cause: blood tests (ANCA, ACE, syphilis serology), urine drug screen where relevant, and biopsy of granulation or of the edge of the perforation (taken from the posterosuperior edge, away from the healed anterior rim).

5.4 Treatment

- **Conservative:** saline or alkaline douches and bland ointment to keep the nose free of crusts.
- **Septal button:** a silicone obturator.
- **Surgical closure:** for small, inactive perforations, with local rotation or advancement flaps and an interposition graft.

CURRENT PRACTICE

The Oxford Textbook reflects a move away from the routine "trial" of a silastic button: some patients find it irritating and obstructive, and a button can enlarge the perforation. Custom-made obturators are more comfortable but lack strong evidence. Perforations larger than about 2 cm are hard to close with local advancement flaps, but techniques borrowed from endoscopic skull base surgery, such as a pedicled flap based on the **anterior ethmoidal artery**, now allow closure of larger defects. Cocaine use should be asked about at any age, and posterior perforations at the bony–cartilaginous junction are the ones that bleed heavily.

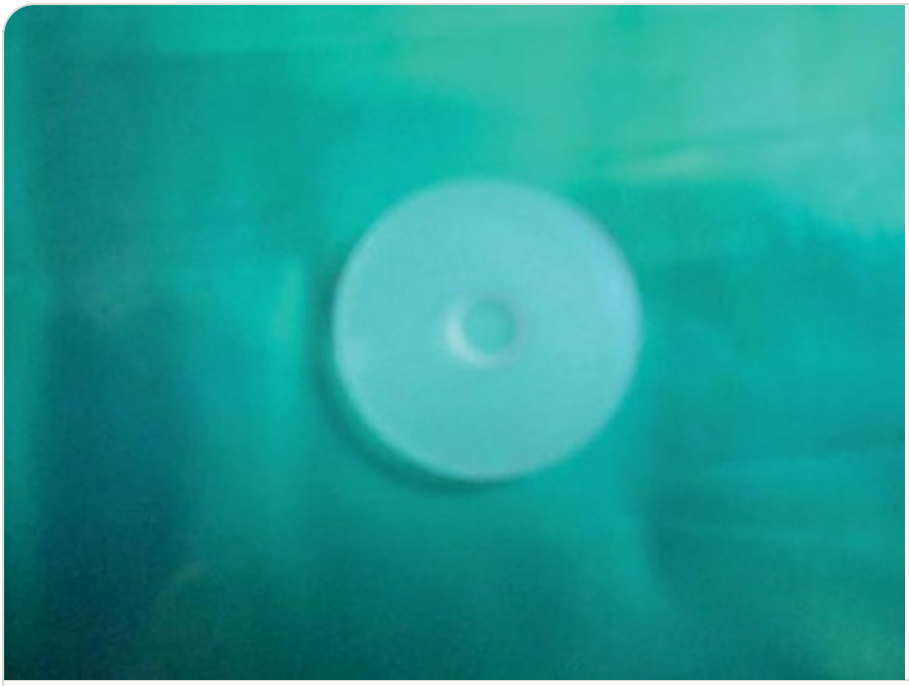


FIGURE 10 A silicone septal button closing a perforation.
Dhingra, fig. 26.12

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 On call

A BLOCKED NOSE AFTER A BLOW

1 Look at the septum in every patient with nasal trauma.

2. A boggy, swollen septum is a haematoma: drain it the same day.
3. Pack both sides and give antibiotics.
4. Review for reaccumulation and abscess.

6.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Parts of the septum** and its framework.
- **Types of deviation,** and the symptoms each causes.
- **Cottle's test** and the nasal valve.
- **Septoplasty versus submucous resection.**
- **Septal haematoma and abscess:** treatment and complications.
- **Causes of septal perforation,** and which diseases perforate the bony or cartilaginous part.

6.3 Common pitfalls

COMMON PITFALLS

- Missing a septal haematoma after trauma.
- Operating on a septum when rhinitis is the real cause of obstruction.
- Biopsying the healed anterior edge of a perforation.

Written for this app from Oxford Textbook of Otolaryngology ch. 28, 33, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 26 and K. J. Lee ch. 31. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

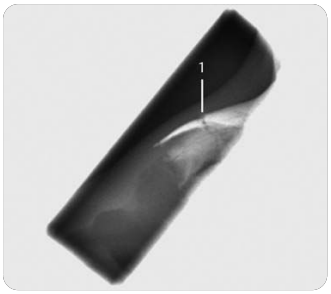
On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · TRAUMA

Fracture of the Nasal Bone and Retention Cysts

- In cases of large cysts resulting in obstruction of the maxillary sinus and those accompanied by signs of sinusitis, surgical removal (marsupialization) might be considered to improve drainage.



Plain radiograph, lateral view. This view of the nasal bone in a patient with facial trauma demonstrates a fracture of the nasal bone (1), with slight…